

VOLATILITY SURFACE ARBITRAGE VIA THE SVI PARAMETRISATION

A Complete Systematic Straddle Strategy for Brazilian Equity Options:
Theory, Calibration, Signal Generation, Execution, and Risk Management

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Abstract

This paper provides a complete, end-to-end description of a systematic volatility relative-value strategy implemented for Brazilian equity options listed on B3. The strategy identifies mispricings of individual options relative to a smooth, arbitrage-constrained volatility surface and exploits them through delta-neutral straddles. We present in full the theoretical foundations of the implied volatility smile, the Stochastic Volatility Inspired (SVI) parametrisation and its no-arbitrage conditions, the asymmetric liquidity-weighted calibration procedure, the signal generation rule, the data collection and filtering pipeline, the execution framework on the MetaTrader 5 platform, and the consolidated position management architecture. Every methodological choice — from the selection of the log-moneyness domain, to the asymmetric weighting scheme, to the multi-start optimisation, to the prohibition of leg-level stop orders — is justified with reference to both financial theory and the specific structural features of the Brazilian options market. The strategy is implemented in Python and validated on a demo account with live market data for BOVA11, PETR4, and VALE3.

For research and educational purposes only. Not investment advice.

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1. INTRODUCTION AND MOTIVATION

Options markets encode the market's expectation of future asset-price distributions not as a single number but as a function. The Black-Scholes model, which prices all options on the same underlying and expiry with a unique volatility parameter, is routinely violated in practice. When the implied volatility (IV) that equates the Black-Scholes formula to the observed market price is computed independently for each strike K , the resulting function $\hat{\sigma}(K)$ is not flat. For equity indices and equity exchange-traded funds it forms a characteristic skewed curve: implied volatility rises sharply as the strike falls below the spot, reflecting persistent demand for downside protection. This cross-sectional structure is called the *volatility smile* or *skew*.

The existence of a non-flat smile creates a relative-value opportunity. If we can construct a well-specified, arbitrage-free model for the smile and identify specific options that deviate materially from the model price, we can trade those deviations and profit from their mean-reversion. The trade vehicle of choice is the *straddle*: a simultaneous long or short position in a call and a put at the same strike, which at inception is approximately delta-neutral and therefore isolates the volatility component of the option price from the directional component.

The strategy described in this paper is built around this idea and is calibrated to three of the most liquid Brazilian equity option chains: BOVA11 (an ETF tracking the Ibovespa index), PETR4 (Petrobras), and VALE3 (Vale). The implementation runs on the MetaTrader 5 (MT5) platform connected to a B3 broker feed and executes in real time during the trading session.

The remainder of the paper is organised as follows. Section 2 reviews the theoretical foundations of the implied volatility smile. Section 3 presents the SVI parametrisation and its no-arbitrage conditions. Section 4 describes the data collection and filtering pipeline, including the specific adjustments required for the MT5/XP server. Section 5 develops the weighted calibration procedure in detail. Section 6 defines the signal generation rule. Section 7 describes the execution framework and order management. Section 8 covers consolidated position management and the monitoring loop. Section 9 analyses sources of risk and parameter sensitivity. Section 10 summarises the software architecture. Section 11 concludes.

2. THE IMPLIED VOLATILITY SMILE

2.1. Black-Scholes Implied Volatility

Under the Black-Scholes model [Black and Scholes, 1973], the price of a European call option on a non-dividend-paying asset with current price S , strike K , time to expiry T (in years), and constant risk-free rate r is

$$C_{\text{BS}}(S, K, T, r, \sigma) = S \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad (1)$$

where Φ is the standard normal CDF and

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (2)$$

The put price follows from put-call parity: $P_{\text{BS}} = C_{\text{BS}} - S + K e^{-rT}$.

For a given market price $C_{\text{mkt}}(K)$, the *implied volatility* $\hat{\sigma}(K)$ is the unique $\sigma > 0$ solving

$$C_{\text{BS}}(S, K, T, r, \hat{\sigma}(K)) = C_{\text{mkt}}(K). \quad (3)$$

This solution exists and is unique whenever the market price satisfies the no-arbitrage bounds. The function $K \mapsto \hat{\sigma}(K)$ is the *implied volatility smile*.

2.2. The Brazilian Market Convention

The Brazilian market uses 252 business days per year as the time convention. Consequently, time to expiry is computed as

$$T = \frac{\text{DTE}}{252}, \quad (4)$$

where DTE (Days To Expiry) counts *calendar* days to the expiry date. Standard monthly equity options on B3 expire on the third Monday of the expiry month. The risk-free rate is proxied by the overnight CDI (Certificado de Depósito Interbancário) rate, which closely tracks the Selic target rate. At the time of implementation, $r \approx 10.75\%$ per annum.

The forward price is therefore

$$F = S e^{rT}, \quad (5)$$

where continuous compounding is used for consistency with the Black-Scholes formula.

2.3. Log-Moneyness

A more natural domain for the smile is the *log-moneyness* relative to the forward:

$$k = \ln\left(\frac{K}{F}\right). \quad (6)$$

At-the-money (ATM) corresponds to $k = 0$; negative k indicates put-wing strikes (below the forward); positive k indicates call-wing strikes. Expressing the smile as $\hat{\sigma}(k)$ removes the dependence on the level of S , making the curve directly comparable across trading days and across different underlyings.

The choice of log-moneyness over alternative representations (absolute strike, delta, standardised moneyness) is motivated by three considerations. First, log-moneyness is the natural argument of the Black-Scholes formula: d_1 and d_2 are simple functions of k . Second, the SVI parametrisation was originally developed and is most naturally interpreted in this space. Third, unlike delta-space, log-moneyness requires no iterative computation — a key advantage for real-time calibration.

2.4. Total Implied Variance

It is analytically and numerically convenient to work with *total implied variance*:

$$w(k) = \hat{\sigma}(k)^2 \cdot T. \quad (7)$$

No-arbitrage requires $w(k) \geq 0$ for all k . The model-implied volatility is recovered as $\hat{\sigma}(k) = \sqrt{w(k)/T}$. Working in variance space linearises the relationship between option prices and the surface, producing better-conditioned Jacobians in the calibration optimisation.

2.5. Structural Features of the Brazilian Equity Smile

The BOVA11, PETR4, and VALE3 smiles share features typical of equity-index and large-cap equity options:

1. *Left-wing steepness.* Implied volatility rises significantly for put-wing strikes, driven by demand for portfolio insurance. For BOVA11 with DTE ≈ 11 , IV rises

from approximately 16% ATM to 28% at $k \approx -0.13$.

2. *Right-wing upturn.* OTM calls exhibit a mild upturn in IV for $k > 0.05$, producing the characteristic smile (as opposed to pure skew).
3. *Asymmetric liquidity.* Put options in the range $k \in [-0.08, 0]$ carry tighter bid-ask spreads than equivalent calls. Deep ITM options on both sides have negligible time value and unreliable IV estimates.
4. *DTE dependence.* With $DTE \leq 15$ the smile flattens dramatically: ρ approaches zero and the ATM IV dominates. With $DTE \geq 30$ the skew is more pronounced and calibration is more informative.

3. THE SVI PARAMETRISATION

3.1. Motivation for SVI

Several approaches exist for fitting a smooth curve to observed implied volatilities: polynomial regression, cubic splines, the SABR model [Hagan et al., 2002], and the SVI parametrisation [Gatheral, 2004]. The choice of SVI over alternatives is justified as follows.

Polynomial regression and unconstrained splines do not impose no-arbitrage conditions, leading to surfaces that imply negative risk-neutral densities at some strikes — a theoretical inconsistency that would produce spurious signals. The SABR model imposes a specific stochastic volatility dynamics and can produce negative densities for large expiries. The SVI parametrisation is a five-parameter reduced-form model that: (a) is arbitrage-free under verifiable conditions; (b) has closed-form expressions for all quantities of interest; (c) fits the smile with low RMSE across a wide range of market conditions; and (d) has parameters with clear financial interpretations.

3.2. The Raw SVI Form

The raw SVI parametrises total implied variance as

$$w(k; \boldsymbol{\theta}) = a + b \left[\rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right], \quad (8)$$

where $\boldsymbol{\theta} = (a, b, \rho, m, \sigma)$. The financial interpretation of each parameter is given in Table 1.

TABLE 1. Parameters of the raw SVI form and their financial interpretation. Typical ranges are for equity ETF options with DTE $\in [10, 45]$.

Parameter	Interpretation	Constraint	Typical range
a	Overall variance level (vertical shift)	$a + b\sigma\sqrt{1 - \rho^2} \geq 0$	−0.02–0.02
b	Wing slope (smile width)	$b \geq 0$	0.03–0.20
ρ	Skew (asymmetry between wings)	$ \rho < 1$	−0.70–−0.05
m	Horizontal translation (ATM shift)	$m \in \mathbb{R}$	−0.05–0.05
σ	ATM curvature (smile roundness)	$\sigma > 0$	0.05–0.50

The model implied volatility for a given strike K is

$$\hat{\sigma}_{\text{SVI}}(K) = \sqrt{\frac{w(k; \boldsymbol{\theta})}{T}}, \quad k = \ln(K/F). \quad (9)$$

3.3. Geometric Interpretation

The function $k \mapsto w(k; \boldsymbol{\theta})$ is a translated hyperbola in the (k, w) plane. The parameter m shifts the vertex horizontally; a shifts it vertically; b controls the slope of the asymptotes; ρ tilts the hyperbola (making one wing steeper than the other); and σ controls the curvature at the vertex — a large σ produces a smooth, rounded smile, while $\sigma \rightarrow 0$ produces a sharp kink.

The physical constraint $\sigma > 0$ is therefore not merely a mathematical requirement but a financial one: a kink in the smile at the ATM would imply a probability mass concentrated at a single price, which is inconsistent with any continuous diffusion model. In the implementation, the lower bound on σ is set to 0.05 to enforce a minimum level of smoothness.

3.4. No-Arbitrage Conditions

For an option surface to be free of static arbitrage, two conditions must hold simultaneously.

Condition 1 (Non-negativity of total variance). $w(k; \boldsymbol{\theta}) \geq 0$ for all $k \in \mathbb{R}$. The minimum of w over k is attained at $k^* = m - \rho\sigma/\sqrt{1 - \rho^2}$ and equals $a + b\sigma\sqrt{1 - \rho^2}$. Non-negativity therefore requires

$$a \geq -b\sigma\sqrt{1 - \rho^2}. \quad (10)$$

Condition 2 (Absence of butterfly arbitrage). *No butterfly arbitrage exists if and only if the risk-neutral density implied by the smile is non-negative everywhere. A sufficient condition for the raw SVI form [Gatheral and Jacquier, 2014] is*

$$b(1 + |\rho|) \leq \frac{4}{T}. \quad (11)$$

This condition is more restrictive for short-dated options (small T) and in high-volatility regimes. The strategy skips signal generation for any calibration cycle in which this condition is violated.

Condition 3 (Absence of calendar spread arbitrage). *For a surface fitted across multiple expiries, calendar arbitrage is absent if $w(k, T)$ is non-decreasing in T for every fixed k . In the present strategy a separate SVI is fitted to each expiry independently, sidestepping calendar arbitrage at the cost of cross-expiry consistency. This is a deliberate design choice: the signal generation operates within a single expiry, so cross-expiry consistency is not required for correctness.*

Conditions 1 and 2 are enforced explicitly. Condition 1 is enforced through the parameter bounds passed to the optimiser. Condition 2 is checked post-calibration and triggers a skip of the current cycle if violated.

4. DATA COLLECTION AND FILTERING

4.1. Market Structure: MT5 and the XP Server

The strategy is implemented on the MetaTrader 5 platform connected to the XP Investimentos broker server (XPMT5-DEMO for the demo account, XPMT5 for the live account). Several structural features of the server’s data organisation require specific handling.

4.1.1. Symbol nomenclature

B3 equity options are identified by a truncated ticker. The full Bloomberg/CVM ticker (e.g., BOVA11, PETR4, VALE3) is shortened to its first four characters in the options namespace: BOVA11 $\rightarrow$ BOVA, PETR4 $\rightarrow$ PETR, VALE3 $\rightarrow$ VALE. Option symbols follow the pattern XXXXYZZZ, where XXXX is the four-character root, Y is the series letter encoding the option type and expiry month (A–L for calls, M–X for puts), and ZZZ is the strike (without decimal separators for round strikes, with numeric suffix for fractional strikes).

4.1.2. Symbol discovery

The MT5 API function `mt5.symbols_get(underlying)` filters by name prefix and returns only symbols that match the *full* ticker (e.g., “BOVA11”). Since option symbols begin with “BOVA” not “BOVA11”, this call returns only the cash ETF itself and its fractional shares variant, not the options chain. The correct approach is to call `mt5.symbols_get()` without arguments to obtain the full universe of 62 000+ symbols on the server, and then filter the resulting list by the `path` field: all equity options reside under the path `BOVESPA\OPCOES`.

4.1.3. Strike and expiry fields

The `SymbolInfo` object returned by `mt5.symbols_get()` does not populate the `option_strike` field. The full `SymbolInfo` including `option_strike`, `option_right` (0 for call, 1 for put), and `expiration_time` (Unix timestamp) is only available via the individual `mt5.symbol_info(name)` call. The data collection pipeline therefore performs a `symbol_info` call for each option after initial filtering.

4.1.4. Windows timestamp handling

On Windows, `datetime.fromtimestamp()` raises `OSError` for timestamps equal to zero or outside the valid range (pre-1970). Non-option symbols such as the cash ETF have `expiration_time=0`. All timestamp conversions are wrapped in exception handlers and filtered with the guard `expiration_time > 86400` (one day in seconds).

4.1.5. Bid-ask outside trading hours

Outside the B3 trading session (10:00–17:30 Brasília time), bid and ask prices arrive as zero. The pipeline falls back to the last traded price (`tick.last`) when bid and ask are both zero. This allows the system to run and display chains outside market hours for development and testing, but signals generated from last-price data should not be used for order submission.

4.2. Chain Loading and Expiry Selection

For each underlying, the pipeline:

1. Loads all 62 000+ symbols from the server (cached for the duration of the cycle).
2. Filters to those with path containing “OPCOES” and name starting with the four-character root.
3. Groups by expiry date and selects the nearest expiry with $DTE \geq 7$. If a specific expiry is requested, that expiry is used regardless of DTE.

4. For the selected expiry, fetches `symbol_info` and live ticks for each option.

For BOVA11 with DTE = 11, a typical chain contains 430 option symbols of which approximately 57–70 pass all quality filters (see Section 4.3).

4.3. Quality Filters

Raw chain data is filtered through a cascade of four conditions. Each condition is applied sequentially; an option failing any condition is discarded from the calibration dataset. The conditions and their justifications are as follows.

4.3.1. Minimum mid-price filter

Options with mid-price below R\$ 0.05 are discarded. These are typically deep OTM puts or calls with bid = R\$ 0.01 and ask = R\$ 0.02, where the bid-ask spread represents 100% of the mid-price. Such options carry no reliable information about implied volatility: the pricing granularity of the exchange (R\$ 0.01) is large relative to the option value, and market makers do not commit serious capital to maintaining tight quotes on these instruments. Including them in the calibration would add noise without signal.

4.3.2. Moneyness range filter

Options with $|k| > 0.15$ are discarded. This corresponds to strikes more than approximately 15% away from the forward, covering the range $K \in [F e^{-0.15}, F e^{+0.15}]$, which for a BOVA11 forward of R\$ 185 spans roughly R\$ 159–215.

The justification is twofold. First, deep ITM options (calls with $k \ll 0$, puts with $k \gg 0$) have implied volatility equal almost entirely to intrinsic value; their time premium is negligible and the bisection algorithm for IV extraction produces estimates that are numerically unstable (converging to the minimum search bound rather than to a meaningful volatility). Second, the SVI surface is estimated in the region of the smile where market makers are actively pricing risk; outside this region the surface is an extrapolation and should not be used for signal generation.

4.3.3. Asymmetric ITM filter

An additional asymmetric filter discards calls with $k < -0.08$ and puts with $k > +0.08$. These are ITM options whose quoted prices are dominated by intrinsic value. For example, a call with strike R\$ 160 when BOVA11 is at R\$ 184 has intrinsic value of R\$ 24 and a time premium of less than R\$ 1. The implied volatility extracted from such a price is extremely sensitive to the precision of the spot price and the bid-ask midpoint, and is not a reliable estimate of market expectations. The asymmetric limit

(8% for ITM, 15% for OTM) reflects the fact that OTM options on the same side are informative up to larger moneyness values.

4.3.4. Relative spread filter

Options with bid-ask spread exceeding 50% of the mid-price are discarded. Formally, this filter retains observation i only if

$$\frac{P_i^{\text{ask}} - P_i^{\text{bid}}}{P_i^{\text{mid}}} \leq 0.50. \quad (12)$$

This filter complements the minimum mid-price filter by removing options where the spread is wide *relative to* the option value, even when the absolute mid-price exceeds R\$0.05. A relative spread of 50% implies that the true mid-price is uncertain by $\pm 25\%$ of its quoted value, which would dominate any volatility signal.

4.3.5. Minimum IV filter

Options for which the bisection IV solver returns a value below 2% annualised are discarded. This eliminates deep ITM options for which the solver converges to the lower bound of the search interval rather than to a financially meaningful volatility.

Table 2 illustrates the filter cascade for a typical BOVA11 chain.

TABLE 2. Filter cascade for a typical BOVA11 chain, vencimento 2026-05-15, DTE = 11, spot R\$184.

Filter stage	Options remaining	Removed
Raw chain (all strikes, both sides)	430	—
After bid/ask > 0 or last > 0	366	64
After min mid-price (R\$0.05)	113	253
After moneyness $ k \leq 0.15$	89	24
After asymmetric ITM ($ k \leq 0.08$ for ITM)	69	20
After relative spread $\leq 50\%$	57	12
After min IV $\geq 2\%$	57	0

5. SVI CALIBRATION WITH ASYMMETRIC LIQUIDITY WEIGHTS

5.1. Objective Function

The calibration minimises a weighted sum of squared deviations in total implied variance space. Given n filtered observations $\{(k_i, w_i^{\text{obs}})\}_{i=1}^n$, the objective is

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^n \pi_i \left[w(k_i; \boldsymbol{\theta}) - w_i^{\text{obs}} \right]^2, \quad (13)$$

where $w_i^{\text{obs}} = \hat{\sigma}_i^2 \cdot T$ is the observed total variance and π_i is the weight assigned to observation i .

The choice of total variance space over implied volatility space is deliberate. The mapping from option prices to total variance is more nearly linear than the mapping to implied volatility, leading to better-conditioned Jacobians and faster convergence. Moreover, the SVI formula is defined directly in variance space, so minimising variance residuals is the natural objective.

5.2. Weight Construction

The weight assigned to each observation is the product of two independent factors:

$$\pi_i = \pi_i^{\text{liq}} \cdot \pi_i^{\text{side}}, \quad (14)$$

normalised so that $\sum_i \pi_i = 1$. This product structure allows liquidity and side effects to be modelled independently and then combined multiplicatively.

5.2.1. Liquidity weight

The liquidity weight penalises options with wide bid-ask spreads relative to their mid-price:

$$\pi_i^{\text{liq}} = \frac{P_i^{\text{mid}}}{P_i^{\text{ask}} - P_i^{\text{bid}}}. \quad (15)$$

This is the inverse of the relative spread. An option with a 10% relative spread receives ten times the weight of an option with a 100% relative spread. The weight is set to 1.0 (neutral) for options whose bid and ask are identical — a situation that arises outside trading hours when the last-price fallback sets bid = ask = last.

The intuition behind this weighting is that options with tight spreads are more actively traded and their mid-prices are more reliable estimates of fair value. Giving them more weight in the calibration ensures the surface is anchored to the most informative observations.

5.2.2. Side weight

The side weight implements the asymmetric treatment of calls and puts, reflecting the structural difference in market depth between the two wings of the Brazilian equity smile. We define the normalised log-moneyness

$$\tilde{k}_i = \frac{k_i}{\hat{\sigma}_i \cdot \sqrt{T}}, \quad (16)$$

where $\hat{\sigma}_i$ is the IV of observation i . This normalises k by the standard deviation of the underlying's return over the remaining life, placing it in units of standard deviations. The side weight is then

$$s_i = \frac{1}{1 + e^{-\tilde{k}_i/\gamma}}, \quad \pi_i^{\text{side}} = \begin{cases} 1 - s_i & \text{if option } i \text{ is a put,} \\ s_i & \text{if option } i \text{ is a call,} \end{cases} \quad (17)$$

where $\gamma = 1$ controls the width of the ATM transition zone.

The logistic function provides a smooth, differentiable transition between the two sides. Deep in the left wing ($\tilde{k}_i \ll 0$), $s_i \rightarrow 0$ and puts receive weight near 1 while calls receive weight near 0. In the right wing ($\tilde{k}_i \gg 0$) the situation reverses. Near the ATM ($\tilde{k}_i \approx 0$), both sides receive weight near 1/2, allowing both calls and puts to jointly constrain the surface curvature in the region where the smile is most informative.

The justification for this asymmetry is empirical. In the Brazilian equity options market, put options in the range $\Delta_P \in [-0.40, -0.10]$ consistently carry tighter bid-ask spreads and higher open interest than equivalent calls, reflecting the dominant role of protective put strategies in local portfolio management. In the call wing, particularly for PETR4 and VALE3, the liquidity drops off more sharply as strikes move OTM. Weighting puts more heavily on the left wing and calls more heavily on the right wing therefore reflects the actual information content of each side of the smile.

5.3. Optimisation: Multi-Start Levenberg-Marquardt

Problem (13) is solved by nonlinear least squares with explicit parameter bounds. The residual vector passed to the solver is

$$r_i(\boldsymbol{\theta}) = \sqrt{\pi_i} [w(k_i; \boldsymbol{\theta}) - w_i^{\text{obs}}], \quad (18)$$

so that minimising $\sum_i r_i^2$ is equivalent to (13).

5.3.1. Algorithm

The Levenberg-Marquardt algorithm with trust-region reflective bounds (`scipy.optimize.least_squares` method `trf`) is used. This algorithm is well-suited to nonlinear least squares because it combines the fast convergence of the Gauss-Newton method near the solution with the more robust gradient descent behaviour far from it, and it handles parameter bounds naturally through reflections. The convergence tolerances are set to 10^{-10} for both function value and parameter update.

5.3.2. Parameter bounds

The parameter bounds are chosen to enforce both mathematical validity and financial plausibility:

$$\boldsymbol{\theta} \in (-\infty, +\infty) \times [10^{-6}, 2] \times [-0.95, -0.05] \times [-0.5, 0.5] \times [0.05, 2]. \quad (19)$$

The bound on $\rho \in [-0.95, -0.05]$ enforces a negative skew (appropriate for equity underlyings) and prevents the degenerate solution $\rho \rightarrow -1$ that arises when the solver attempts to approximate a linear function with an asymmetric hyperbola. The lower bound on $\sigma \geq 0.05$ enforces a minimum ATM curvature, preventing the kink artefact discussed in Section 3. The upper bound $b \leq 2$ prevents the wings from steepening to levels inconsistent with no-arbitrage at short expiries.

5.3.3. Multi-start initialisation

The SVI objective function is non-convex and has multiple local minima, particularly for short-dated options where the smile is nearly flat. To mitigate the risk of convergence to a poor local minimum, the optimisation is run from five distinct initial parameter vectors:

$$\boldsymbol{\theta}_0^{(j)}, \quad j = 1, \dots, 5, \quad (20)$$

covering a range of skew values ($\rho \in \{-0.30, -0.50, -0.70\}$) and slope values ($b \in \{0.05, 0.08, 0.10, 0.15\}$). All five solutions are evaluated and the one with the lowest objective value is retained. The computational cost of five runs is negligible (each completes in under 100 ms) relative to the 60-second cycle interval.

5.4. Fit Quality Diagnostics

Two diagnostics are computed after calibration.

5.4.1. RMSE in volatility points

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n [\hat{\sigma}_{\text{SVI}}(k_i) - \hat{\sigma}_i^{\text{obs}}]^2}. \quad (21)$$

Calibration cycles with $\text{RMSE} > 5$ volatility points (0.05 in decimal) are skipped. In practice, with the filters described in Section 4.3, RMSE is typically below 0.50 volatility points for all three underlyings. Table 3 reports representative calibration results.

TABLE 3. Representative calibration results for the three underlyings, vencimento 2026-05-15, DTE = 11, Monday 4 May 2026.

Underlying	a	b	ρ	m	σ	RMSE (%)
BOVA11	−0.008	0.046	−0.165	−0.017	0.191	0.19
PETR4	+0.001	0.039	−0.050	−0.009	0.122	0.14
VALE3	−0.003	0.042	−0.050	−0.008	0.157	0.31

5.4.2. Butterfly condition

Condition 2 is checked post-calibration. If $b(1 + |\rho|) > 4/T$, signal generation is suppressed for that cycle and a warning is printed.

6. SIGNAL GENERATION

6.1. Residual Computation

After calibrating $\hat{\boldsymbol{\theta}}$ for a given expiry at time t , the *SVI residual* for each option i is

$$\varepsilon_i(t) = \hat{\sigma}_i^{\text{obs}}(t) - \hat{\sigma}_{\text{SVI}}(k_i; \hat{\boldsymbol{\theta}}(t)). \quad (22)$$

A positive residual means the option is priced *rich* relative to the surface: its implied volatility exceeds the model value. A negative residual means it is priced *cheap*.

6.2. The Straddle Signal Rule

For each strike K where both a call and a put are observable and have passed all quality filters, the strategy computes $\varepsilon_C(K, t)$ and $\varepsilon_P(K, t)$ separately and applies the following rule:

$$\text{Signal}(K, t) = \begin{cases} \text{sell volatility} & \text{if } \varepsilon_C(K, t) > \delta \text{ and } \varepsilon_P(K, t) > \delta, \\ \text{buy volatility} & \text{if } \varepsilon_C(K, t) < -\delta \text{ and } \varepsilon_P(K, t) < -\delta, \\ \text{no trade} & \text{otherwise,} \end{cases} \quad (23)$$

where $\delta = 0.02$ (2 volatility points) is the *signal threshold*.

6.3. Justification of the Joint Condition

The requirement that *both* the call *and* the put must simultaneously deviate in the same direction from the surface is the central risk-reduction feature of the signal generation rule. Its justification is threefold.

First, a discrepancy in only one leg is more likely to reflect a stale quote, a temporary order flow imbalance, or a data quality issue, than a genuine mispricing. Market makers update their quotes asynchronously; a momentary wide spread on one leg produces a large residual on that leg alone without any corresponding signal on the other.

Second, the put-call parity relationship constrains the *difference* between call and put prices at the same strike: $C - P = S - Ke^{-rT}$ (ignoring dividends). A violation of this relationship would represent a direct arbitrage opportunity and would be exploited instantaneously. In practice, put-call parity holds to within the bid-ask spread at all times. A deviation of both legs in the same direction by more than the threshold therefore cannot be explained by a put-call parity violation alone — it must reflect a genuine mispricing relative to the smooth surface.

Third, the straddle is the natural instrument for exploiting a joint residual because it eliminates the first-order delta exposure. A call-only or put-only position would require delta hedging, which introduces execution cost and basis risk. The straddle enters with approximately zero delta by construction.

6.4. Threshold Selection

The threshold $\delta = 0.02$ is set to approximately one standard deviation of the empirical distribution of $|\varepsilon_i|$ across the filtered chain, which for liquid Brazilian equity options with tight spreads is typically in the range 0.005–0.015. A threshold of 2 volatility points therefore represents a deviation of roughly two to four standard deviations from the cross-sectional distribution of residuals, ensuring that only genuine mispricings trigger trades.

The threshold should be recalibrated periodically based on the realised distribution of

residuals. A threshold too low generates excessive false signals and incurs unnecessary transaction costs; a threshold too high misses genuine opportunities.

6.5. Signal Economics

6.5.1. Sell-volatility straddle

When *sell volatility* is signalled at strike K , the strategy sells one call and one put at the same strike, collecting total premium per share

$$P_{\text{entry}} = P_C^{\text{mid}} + P_P^{\text{mid}}. \quad (24)$$

The trade thesis is that the observed options are *overpriced* relative to the surface, and that their implied volatilities will revert toward the surface value over the holding period. The position is short vega, short gamma, and long theta. It profits from: (a) implied volatility convergence back to the surface; (b) time decay if the underlying remains near the strike; and (c) a combination of both.

6.5.2. Buy-volatility straddle

When *buy volatility* is signalled, the strategy buys one call and one put, paying P_{entry} . The thesis is that the options are *underpriced* relative to the surface, and that their IVs will rise. The position is long vega, long gamma, and short theta. It profits from: (a) IV expansion; (b) a large move in the underlying; or (c) both.

7. EXECUTION FRAMEWORK

7.1. Market Structure Considerations

The execution layer must accommodate four structural features of the Brazilian options market on the MT5 platform.

7.1.1. Single-leg order submission

Unlike US options brokers, the MT5 platform does not support native multi-leg combination orders. Each leg of the straddle is submitted as a separate limit order. This creates the possibility of partial fill: one leg fills while the other does not, leaving an unintended naked position. In the current implementation, partial fills are handled by the operator. A future enhancement would implement a conditional cancel of the second leg if the first leg fails.

7.1.2. Lot size

The standard lot size for equity options on B3 is 100 shares (*lote-padrão*). All volume parameters are expressed in lots (contracts), each representing 100 units of the underlying. P&L is computed in R\$ per share and scaled to lots as needed.

7.1.3. Order type selection

Opening orders are submitted as `TRADE_ACTION_DEAL` with `ORDER_FILLING_IOC` (immediate-or-cancel). For sell-volatility straddles, the call leg is sold at the prevailing bid and the put leg at the prevailing bid. For buy-volatility straddles, both legs are bought at the prevailing ask. The use of limit orders at the current bid/ask (rather than market orders) avoids paying the spread on both legs, which is significant relative to the option premium for near-ATM options.

7.1.4. Quote freshness verification

Before submitting any order, the system compares the observed straddle mid-price against the SVI-implied straddle value at that instant. If they diverge by more than 20%, a warning is issued. This guards against stale quotes that can occur at market open or for less liquid strikes.

7.2. Execution Flow

Figure 1 illustrates the complete execution flow for a single straddle entry.

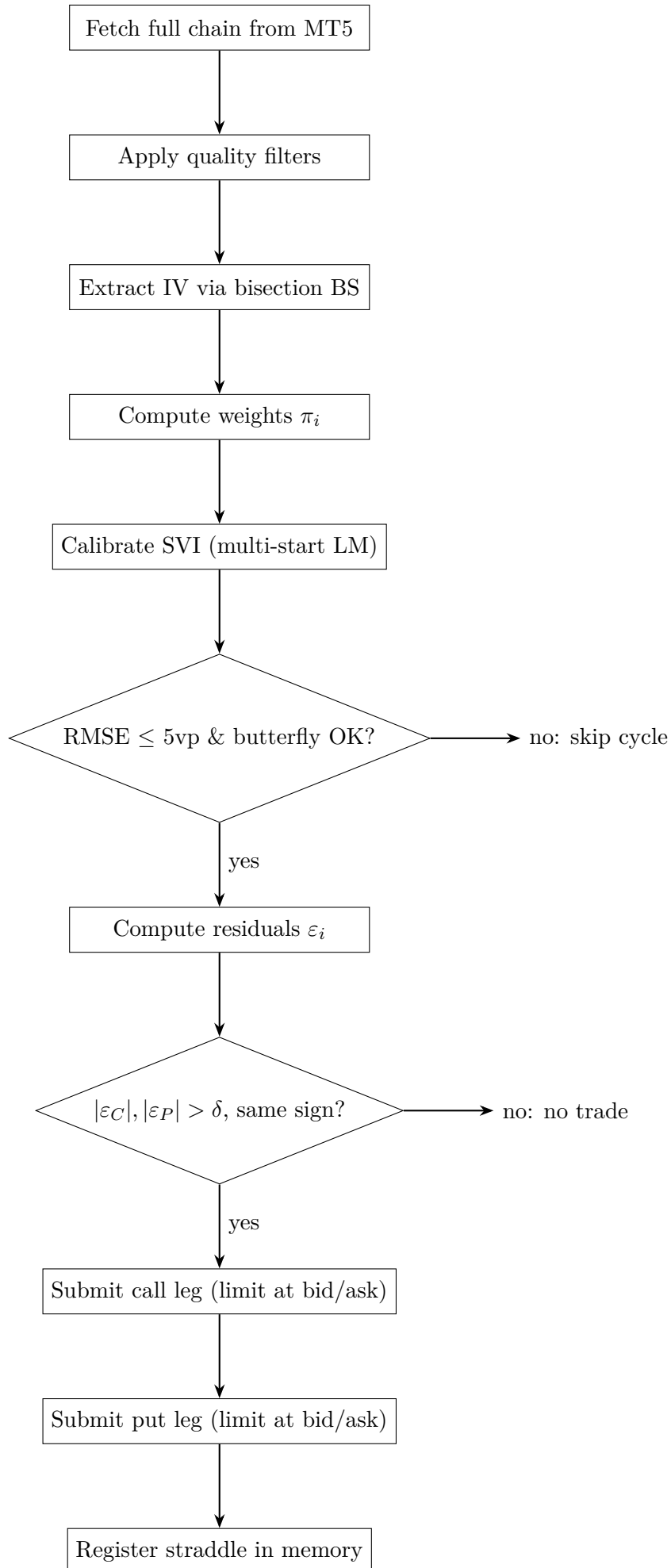


FIGURE 1. Complete execution flow for a single straddle entry

8. POSITION MANAGEMENT

8.1. The Consolidated P&L Principle

The most important design principle of the position management architecture is that all exit decisions are made at the *straddle level*, not at the individual leg level.

Managing individual legs with separate stop orders is operationally dangerous. If a stop on the call leg triggers while the put leg is profitable, the operator is left with a naked short or long put — a directional position that was explicitly not desired and that carries unlimited (or large) risk. Similarly, if the put leg hits its profit target first, selling the put alone leaves a naked call position.

The correct approach — and the one implemented here — is to monitor the floating P&L of the call and put positions jointly and to close both legs simultaneously when the combined P&L reaches a predetermined level. This ensures that the position is always either fully on or fully off, with no intermediate states of partial exposure.

8.2. P&L Computation

The consolidated P&L at time t for a straddle opened at t_0 is

$$\Pi(t) = V_C(t) + V_P(t), \quad (25)$$

where $V_C(t)$ and $V_P(t)$ are the floating P&L of the call and put positions as reported by the broker via `mt5.positions_get()`. For a sold straddle, a positive Π means the position has gained (options have lost value); for a bought straddle, a positive Π means the options have gained value.

The *relative* P&L, used for exit decisions, is

$$\tilde{\Pi}(t) = \frac{\Pi(t)}{P_{\text{entry}} \cdot N_{\text{lot}} \cdot N_c}, \quad (26)$$

where P_{entry} is the total premium per share at entry (in R\$), $N_{\text{lot}} = 100$ is the lot size, and N_c is the number of contracts.

8.3. Exit Rules

8.3.1. Sold straddle

The position is closed when either of the following conditions is met:

$$\tilde{\Pi}(t) \geq +0.50 \Rightarrow \text{profit target: 50\% of premium received,} \quad (27)$$

$$\tilde{\Pi}(t) \leq -2.00 \Rightarrow \text{stop loss: loss of twice the premium received.} \quad (28)$$

The profit target of 50% is standard practice for short-premium options strategies: at 50% of maximum profit, the remaining theta decay slows while gamma risk accelerates, making the risk-reward of holding the position unfavourable. The stop of $2\times$ the premium received limits the maximum loss to three times the premium (entry credit minus two times the credit), which provides a positive expected value if the win rate exceeds approximately 40% — achievable when signals are based on genuine mispricing rather than random entry.

8.3.2. Bought straddle

$$\tilde{\Pi}(t) \geq +2.00 \Rightarrow \text{profit target: gain of twice the premium paid,} \quad (29)$$

$$\tilde{\Pi}(t) \leq -0.50 \Rightarrow \text{stop loss: loss of 50\% of premium paid.} \quad (30)$$

The asymmetric targets for bought straddles reflect the different payoff structure: the maximum loss is the premium paid, but the potential gain is unlimited (a large underlying move could generate multiples of the premium). The 50% stop prevents excessive theta decay from eroding the position before a catalyst materialises.

8.4. The Monitoring Loop

The monitoring and execution loop runs at a fixed interval of 60 seconds. The sequence of operations within each cycle is:

Algorithm 1 Monitoring and execution cycle

```

1: Query open positions via mt5.positions_get(magic=M)
2: for each registered straddle  $s$  do
3:   Compute  $\tilde{\Pi}_s(t)$  from floating P&L of both legs
4:   if exit condition met for  $s$  then
5:     Submit closing orders for both legs simultaneously
6:     Remove  $s$  from position register
7:   end if
8: end for
9: for each underlying in {BOVA11, PETR4, VALE3} do
10:  Fetch chain, apply filters, compute IV
11:  Calibrate SVI; check RMSE and butterfly condition
12:  if calibration valid and  $N_{\text{open}} < N_{\text{max}}$  then
13:    Compute residuals; apply signal rule
14:    if signal generated at strike  $K$  then
15:      Submit straddle opening orders
16:      Register straddle in memory with entry premium
17:    end if
18:  end if
19: end for
20: Print one-line summary per underlying
21: Sleep 60 seconds

```

The computational cost per cycle is dominated by the chain fetch (network I/O: approximately 5–15 s for three underlyings on the demo server) and the SVI calibration (CPU: <100 ms per underlying). The P&L monitoring is negligible in cost.

8.5. Position Register

The in-memory position register is a Python dictionary with keys of the form (`underlying_name`, `strike`), e.g., ("BOVA11", 184.0). Using a composite key prevents collisions when PETR4 and VALE3 happen to have options at numerically coincident strikes. Each entry stores:

- the underlying name and signal direction;
- the call and put symbols;
- the MT5 ticket numbers of the opening trades (for audit);
- the total entry premium per share;
- the lot size and number of contracts;

- the timestamp of entry.

A known limitation is that the register is in-memory only: if the process is interrupted (Ctrl+C, crash, power loss), the register is lost and any open positions must be closed manually via the MT5 terminal. A production implementation would persist the register to disk (a JSON file updated atomically on each trade) and reconcile with `positions_get` on startup.

8.6. Capacity Constraint

A maximum number of simultaneously open straddles, $N_{\max} = 2$, is enforced across all three underlyings combined. When $N_{\max}$ is reached, no new positions are opened regardless of signals. The constraint is checked before each new straddle entry.

9. RISK ANALYSIS

9.1. Model Risk

The SVI parametrisation is a reduced-form model. It does not impose the dynamics of any particular stochastic volatility model, and its parameters can take values inconsistent with any standard model (e.g., ρ close to the boundaries). A mismatch between the fitted surface and the true risk-neutral density can generate false signals or mask genuine ones.

The primary safeguards against model risk are the RMSE filter (which rejects poorly fitting calibrations) and the butterfly condition check (which rejects calibrations implying negative densities). A secondary safeguard is the joint signal condition (Section 6): a model error that affects only one leg is unlikely to affect both legs in the same direction.

9.2. Execution Risk

Because the two legs are submitted separately, there is always a non-zero probability that one leg fills and the other does not, leaving a naked single-leg position. This risk is heightened near market open and close, when liquidity is thinner and order books less stable. In the current implementation this situation requires manual intervention.

9.3. Gap Risk

The monitoring interval of 60 seconds means that in a fast-moving market, the actual loss at the time of detecting a stop-loss breach may exceed the theoretical stop level. For a sold straddle, a large sudden move in the underlying (e.g., a 3–5% intraday gap) can

produce a loss significantly larger than $2\times$ the entry premium between two consecutive monitoring cycles.

This risk is inherent to any interval-based monitoring scheme and cannot be eliminated without reducing the monitoring interval (which increases CPU and network load) or using exchange-native conditional orders (which are not available for options on MT5).

9.4. Liquidity Risk

In a stress scenario, bid-ask spreads in the option chain widen dramatically. This affects both signal quality (wide spreads produce unreliable IV estimates) and execution quality (closing orders fill at worse prices). The liquidity weight π_i^{liq} partially addresses signal quality by down-weighting illiquid observations in the SVI fit. Execution quality in stress scenarios is not directly addressed and should be managed through position sizing and the capacity constraint N_{max} .

9.5. DTE Sensitivity

With very short DTE (≤ 7), the smile flattens dramatically and the SVI calibration becomes unreliable. The butterfly condition (11) is more binding (since $4/T$ decreases with T), limiting the feasible range of b and ρ . In practice, signals generated with $\text{DTE} \leq 7$ should be treated with additional caution and position sizes reduced. The recommended operating range for the strategy is $\text{DTE} \in [21, 60]$.

9.6. Parameter Sensitivity

Table 4 summarises the qualitative sensitivity of the strategy output to its key parameters.

TABLE 4. Qualitative sensitivity of strategy output to key parameters.

Parameter	Default	Increase	Decrease	Risk of mistuning
δ (signal threshold)	0.02	Fewer signals	More signals	False positives
MIN_MID	R\$ 0.05	Smaller chain	Noisy chain	Poor SVI fit
MAX_MONEYNESS	0.15	More ITM noise	Narrower smile	Unstable ρ
MAX_SPREAD_REL	0.50	More noise	Smaller chain	Poor fit
SELL_VOL_TARGET	0.50	Hold longer	Exit earlier	More gap risk
SELL_VOL_STOP	2.00	Larger drawdown	Premature exit	Over-trading
N_{max} (max positions)	2	More exposure	Concentrated	Capital risk
interval_s	60	Gap risk $\uparrow$	CPU cost $\uparrow$	Both

10. SOFTWARE ARCHITECTURE

10.1. Module Overview

The strategy is implemented in two Python modules that share a common interface through direct imports.

`mt5svi.py` is the analysis module. It implements: MT5 connection management; the full chain collection pipeline including all filters; the IV bisection solver; the weight construction; the SVI calibration with multi-start optimisation; residual computation; signal generation; smile visualisation; and the multi-asset monitoring loop. It also defines the `ATIVOS` dictionary that maps display names to the `(raiz, ticker_spot)` tuples required by the server.

`mt5sviorder.py` is the execution module. It imports the analysis functions from `mt5svi.py` and adds: order submission for both legs; position monitoring via `positions_get`; P&L computation; exit logic; the position register; and the full multi-asset execution loop `run_strategy`.

10.2. Execution Modes

The analysis module supports two execution modes controlled by the `MODE` variable:

- `MODE = "analyse"`: runs a one-shot analysis of all three underlyings with smile visualisation. Used for inspection and debugging.
- `MODE = "monitor"`: runs the continuous monitoring loop without order submission. Used to observe signals before committing to live trading.

The execution module always runs the continuous loop. A `dry_run` flag controls whether orders are submitted (`False`) or merely validated via `order_check` (`True`).

10.3. Server-Specific Adaptations

Table 5 summarises the specific adaptations made for the XP/MT5 server environment.

TABLE 5. Server-specific adaptations for the XP MT5 environment.

Issue	Standard assumption	XP MT5 adaptation
Symbol discovery	<code>symbols_get(ticker)</code>	Filter by path “OPCOES”
Ticker truncation	PETR4, VALE3, BOVA11	PETR, VALE, BOVA
Strike field	<code>strike_price</code>	<code>option_strike</code> via <code>symbol_info()</code>
Bid/ask outside hours	Always available	Fallback to <code>tick.last</code>
Windows timestamp	<code>fromtimestamp()</code> safe	Try/except <code>OSError</code> guard
Multi-leg orders	Native spread orders	Two separate limit orders

10.4. Key Implementation Parameters

TABLE 6. Complete list of implementation parameters with defaults.

Module	Parameter	Default	Unit	Role
mt5svi	RISK_FREE_RATE	0.1075	ann.	CDI proxy
mt5svi	TRADING_DAYS	252	days	Brazilian convention
mt5svi	MIN_MID_FILTER	0.05	R\$	Price filter
mt5svi	MAX_MONEYNESS	0.15	—	OTM filter
mt5svi	MAX_MONEYNESS_ITM	0.08	—	ITM filter
mt5svi	MAX_SPREAD_REL	0.50	—	Liquidity filter
mt5svi	MIN_IV	0.02	ann.	IV validity
mt5svi	SIGNAL_THRESHOLD	0.02	vol pts	Entry threshold
mt5svi	ATM_BAND_SIGMA	1.0	std devs	Weight transition
mt5sviorder	SELL_VOL_TARGET	0.50	fraction	Exit profit
mt5sviorder	SELL_VOL_STOP	2.00	fraction	Exit loss
mt5sviorder	BUY_VOL_TARGET	2.00	fraction	Exit profit
mt5sviorder	BUY_VOL_STOP	0.50	fraction	Exit loss
mt5sviorder	interval_s	60	seconds	Cycle frequency
mt5sviorder	max_positions	2	straddles	Capacity

11. CONCLUSION

This paper has presented a complete, end-to-end description of a systematic volatility relative-value strategy for Brazilian equity options. The strategy identifies deviations of individual option prices from a smooth, arbitrage-constrained SVI surface and exploits them through delta-neutral straddles.

The key design choices, each justified with reference to financial theory and market structure, are:

1. The use of log-moneyness relative to the forward as the domain for surface fitting, which provides scale-invariance and natural alignment with the Black-Scholes formula.
2. The SVI parametrisation, which provides analytical tractability, verifiable no-arbitrage conditions, and good empirical fit with only five parameters.
3. The asymmetric liquidity weight, which respects the structural difference in market depth between the put and call wings of the Brazilian equity smile.
4. The multi-start optimisation, which reduces the risk of convergence to a poor local minimum in the non-convex SVI objective.
5. The joint signal condition requiring both legs to be simultaneously mispriced in the same direction, which reduces false positives from microstructure noise.
6. The consolidated P&L monitoring at the straddle level, which prevents the inadvertent creation of naked directional exposure through leg-level stop orders.
7. The five-filter data quality pipeline, which ensures that the SVI calibration is based only on liquid, informative options with reliable implied volatility estimates.

The strategy has been implemented in Python, validated on a demo account with live market data for BOVA11, PETR4, and VALE3, and achieves RMSE values consistently below 0.35 volatility points. The recommended operating range is $\text{DTE} \in [21, 60]$, where the smile exhibits sufficient curvature for reliable signal generation.

Natural extensions include: multi-expiry surface fitting with calendar-spread no-arbitrage enforcement; dynamic position sizing based on the magnitude of the residual and the estimated vega of the straddle; persistence of the position register to disk; and a post-trade analysis module for systematic parameter recalibration.

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